EXPERIMENT 1

Student Name: Ridhima Gulati **UID: 24BCY70132**
Branch: AIT-CSE (Cyber Security) **Section/Group: 24AIT-KRG-G2**
Semester: 5TH
Subject Name: Computer Network **Subject Code: 24CSH-337**

Experiment 1 – Understanding Networking Fundamentals using Cisco Packet Tracer

Aim

To understand the fundamentals of computer networking by studying IP addressing, Cisco IOS, Ethernet cables, RJ-45 connectors, Layer-2 switches, and routers, and to demonstrate their working using Cisco Packet Tracer.

Objective

- To understand the concept of IP addressing and subnet masks.
- To study the Cisco IOS Command Line Interface (CLI).
- To understand the difference between Straight-through and Cross-over cables.
- To identify the purpose of the RJ-45 connector.
- To understand the working of a Layer-2 Switch.
- To understand the role of a Router in connecting different networks.
- To verify network communication using the Ping command.

Software Requirements

Software

- Cisco Packet Tracer
- Windows Operating System

Hardware (Virtual)

- PC-PT
- Cisco 2960-24TT Switch
- Cisco 1941 Router

Practical / Experiment Steps

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Part I – IP Address

1. Place two PCs and one Layer-2 Switch in Cisco Packet Tracer.
2. Connect the PCs to the switch using Copper Straight-Through cables.
3. Assign IP addresses to both PCs.
4. Verify connectivity using the Ping command.

Part II – Cisco IOS

5. Add a Cisco 1941 Router.
6. Access the CLI.
7. Explore User EXEC Mode, Privileged EXEC Mode, and Global Configuration Mode.
8. Execute basic Cisco IOS commands.

Part III – Straight Cable & Cross Cable

9. Demonstrate Straight-through cable connections.
10. Demonstrate Cross-over cable connections.
11. Observe the difference between both cable types.

Part IV – RJ-45 Connector

12. Study the RJ-45 Ethernet connector and its purpose in networking.

Part V – Layer-2 Switch

13. Connect multiple PCs to a Layer-2 Switch.
14. Verify communication between all connected hosts.

Part VI – Router

15. Connect two different networks using a router.
16. Configure router interfaces.
17. Verify communication between the two networks.

Procedure

IP Address

1. Open Cisco Packet Tracer.

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2. Add two PCs and one Switch.
3. Connect using Copper Straight-Through cable.
4. Configure:

PC0

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0

PC1

- IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.0

5. Open Command Prompt.
6. Execute:

ping 192.168.1.20

Cisco IOS

1. Add a Cisco 1941 Router.
2. Open CLI.
3. Type:

enable

4. Execute:

show running-config

show startup-config

show ip interface brief

5. Enter Global Configuration Mode.

configure terminal

6. Change hostname.

hostname R1

7. Save configuration.

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copy running-config startup-config

Straight Cable

1. Connect PC → Switch using Copper Straight-Through cable.
2. Connect Switch → Router using Copper Straight-Through cable.

Cross Cable

1. Connect PC → PC using Copper Cross-Over cable.
2. Connect Switch → Switch using Copper Cross-Over cable.

Layer-2 Switch

1. Connect three PCs to a Layer-2 Switch.
2. Assign IP addresses.
3. Verify communication using Ping.

Router

1. Connect two switches to a router.
2. Configure router interfaces.
3. Configure default gateways on PCs.
4. Verify communication using Ping.

Input / Output Details

Devices Used

- PC-PT
- Cisco 2960-24TT Switch
- Cisco 1941 Router

Cable Types

- Copper Straight-Through
- Copper Cross-Over

Cisco IOS Commands Used

enable

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configure terminal

hostname R1

show running-config

show startup-config

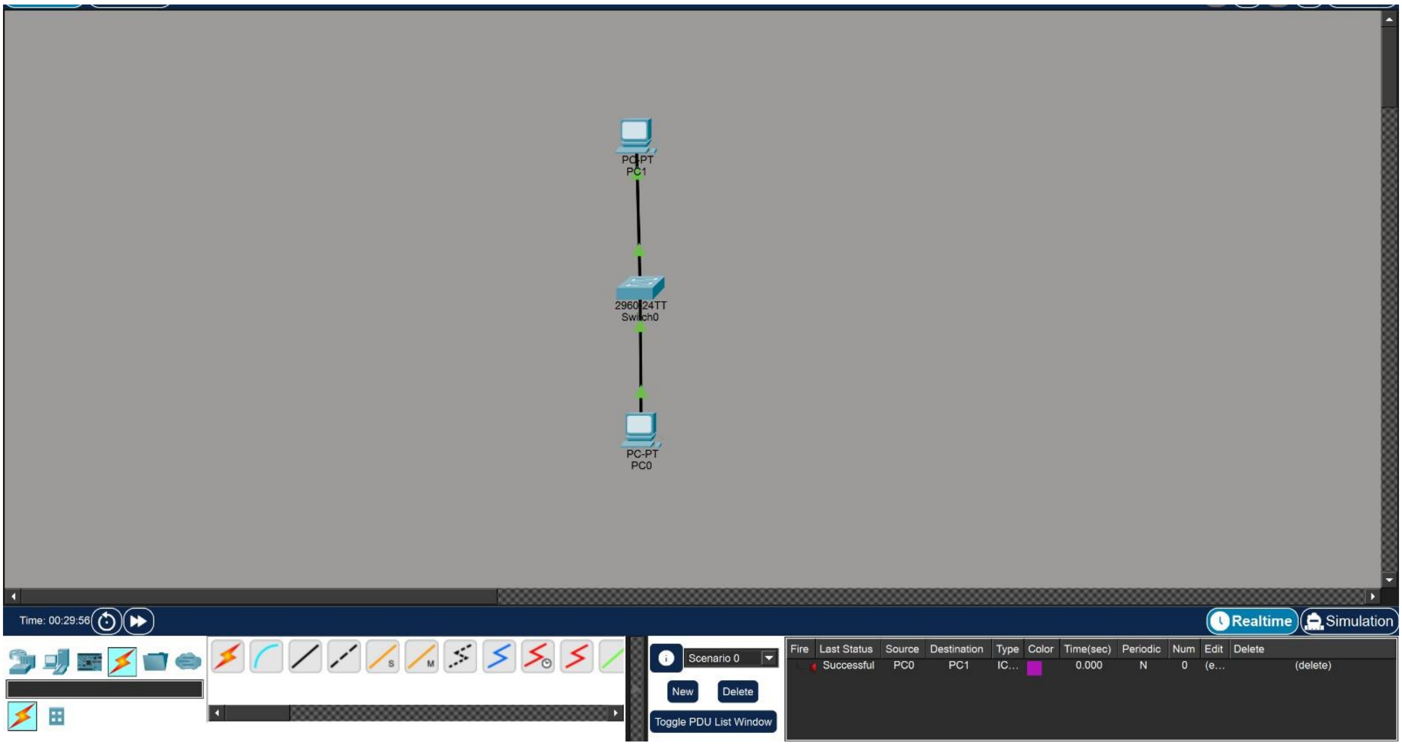
show ip interface brief

copy running-config startup-config

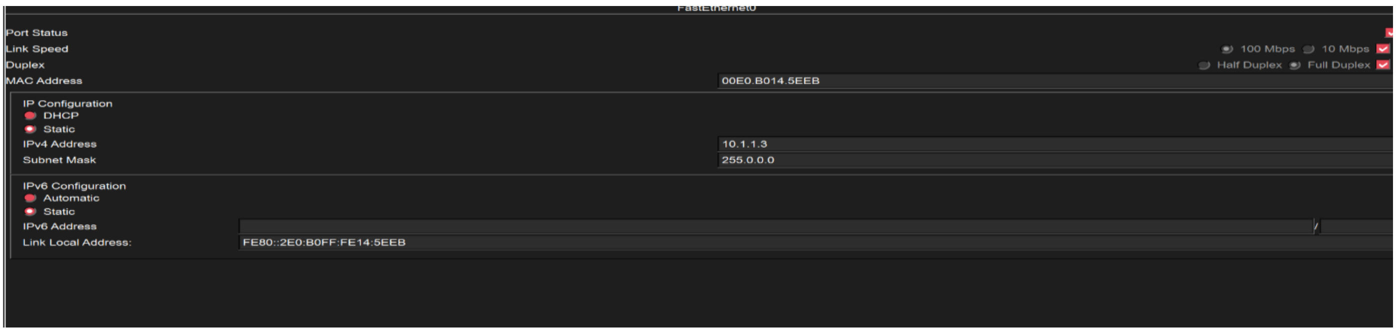
ping

Step-wise Output

Step 1 – Place Two PCs and One Switch And Connect Them



Step 2 – Configure IP Address of PC0



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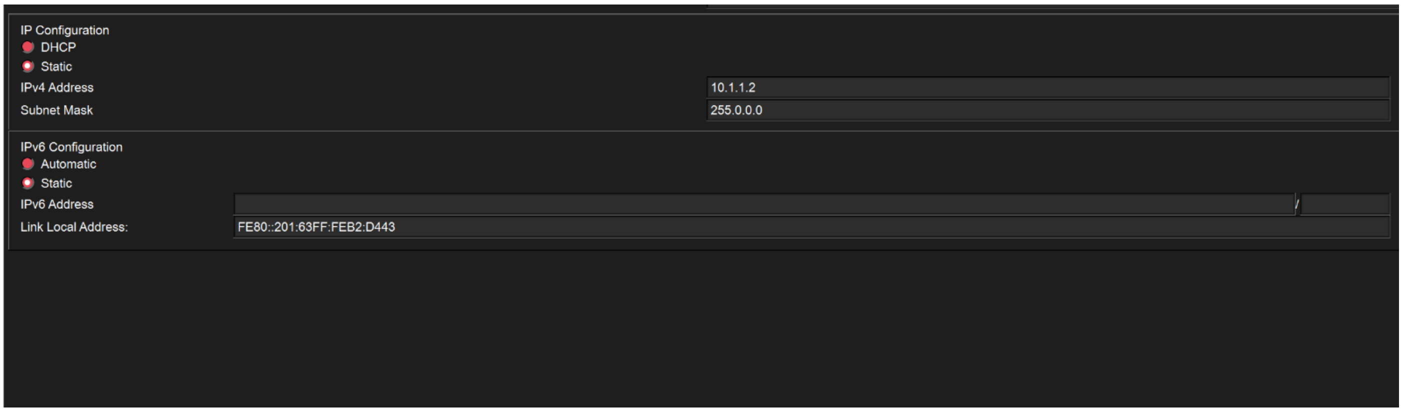
Step 3 – Configure IP Address of PC1



Step 5 – Verify Communication Using Ping

```
C:\>ping 10.1.1.3

Pinging 10.1.1.3 with 32 bytes of data:

Reply from 10.1.1.3: bytes=32 time<1ms TTL=128
Reply from 10.1.1.3: bytes=32 time<1ms TTL=128
Reply from 10.1.1.3: bytes=32 time<1ms TTL=128
Reply from 10.1.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time=4ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time=3ms TTL=128
Reply from 10.1.1.2: bytes=32 time=4ms TTL=128

Ping statistics for 10.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms
```

Step 6 – Add Cisco 1941 Router

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Step 7 – User EXEC Mode (Router>)

```
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>show clock
*0:0:37.482 UTC Mon Mar 1 1993
Router>show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 23-Feb-11 14:19 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco1941 uptime is 43 seconds
System returned to ROM by power-on
System image file is "flash0:c1900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
 --More-- |
```

Step 8 – Privileged EXEC Mode (Router#)

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```
Router#show startup-config
startup-config is not present
Router#show ip interface
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#show history
  show clock
  show version
  enable
  show running-config
  show startup-config
  show ip interface brief
  show history
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#banner motd #Welcome#
R1(config)#enable secret 123
R1(config)#service password-encryption
R1(config)#
```

Step 9 – Global Configuration Mode (Router(config)#)

```
  show history
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#banner motd #Welcome#
R1(config)#enable secret 123
R1(config)#service password-encryption
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

Step 11 – Straight Cable Demonstration

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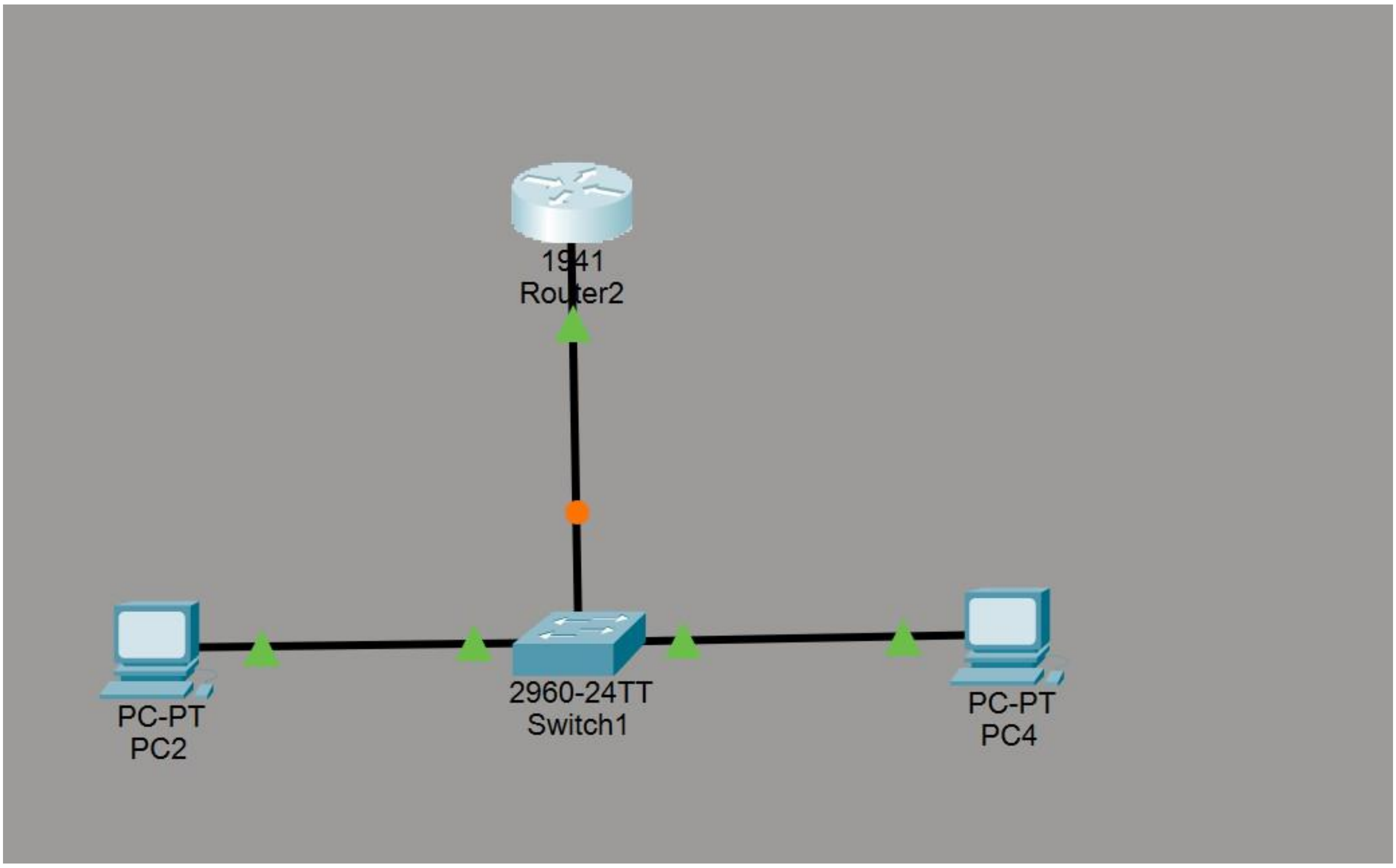
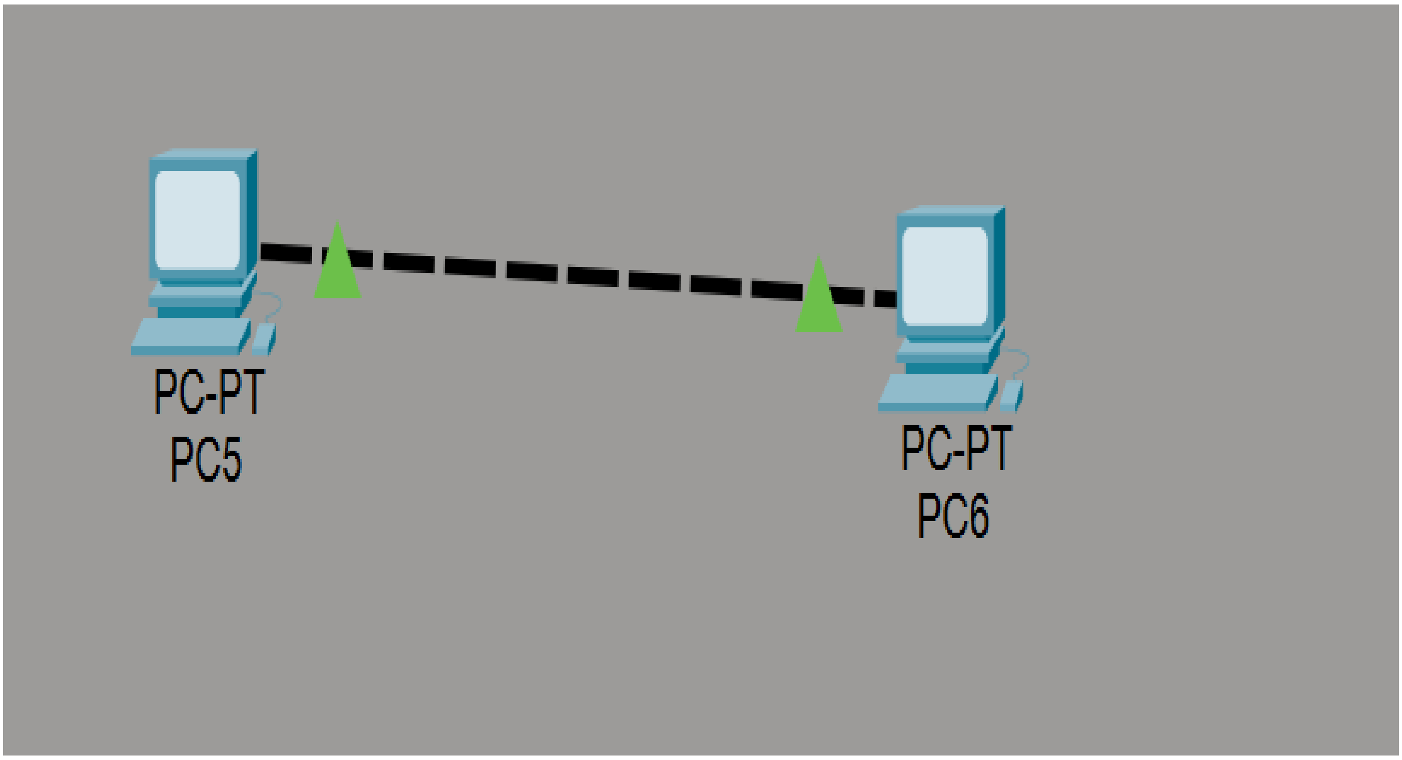


Step 12 – Cross Cable Demonstration



Step 13 – Layer-2 Switch Demonstration

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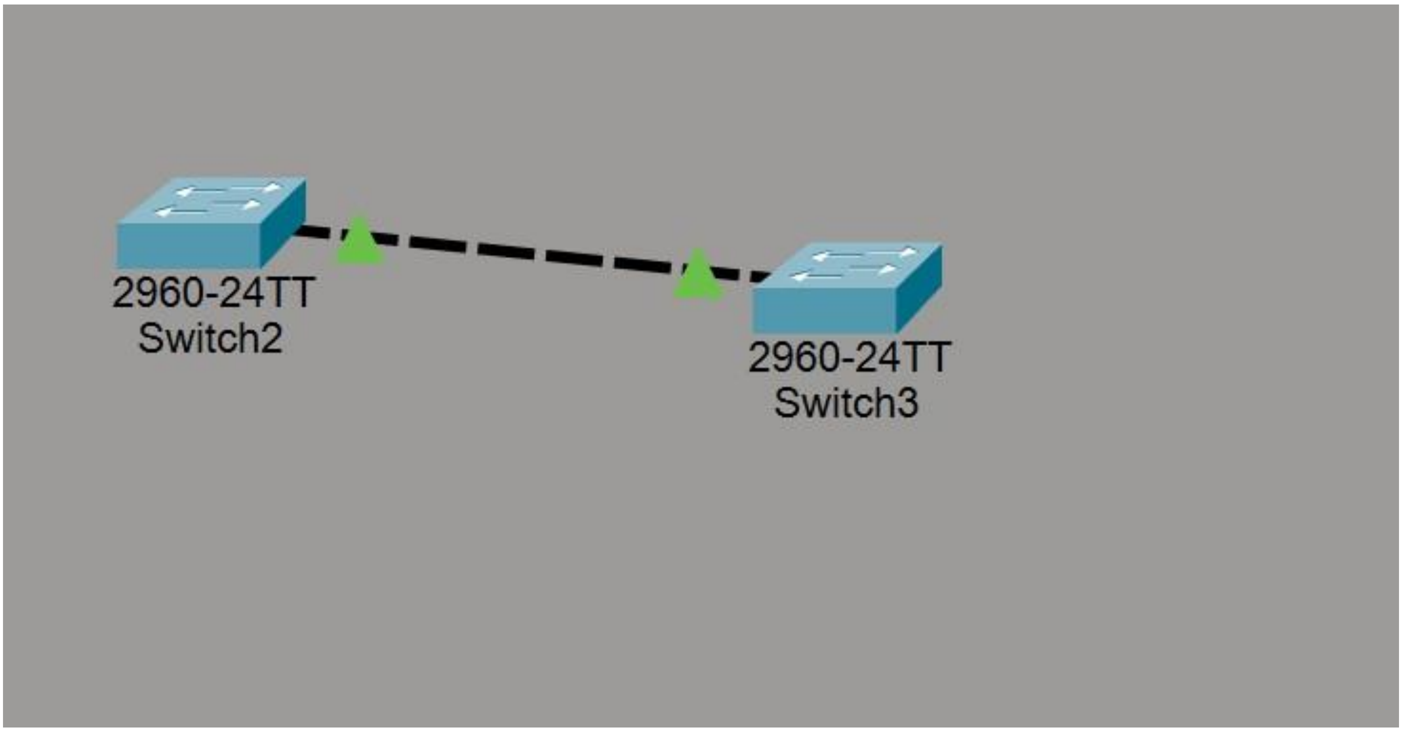
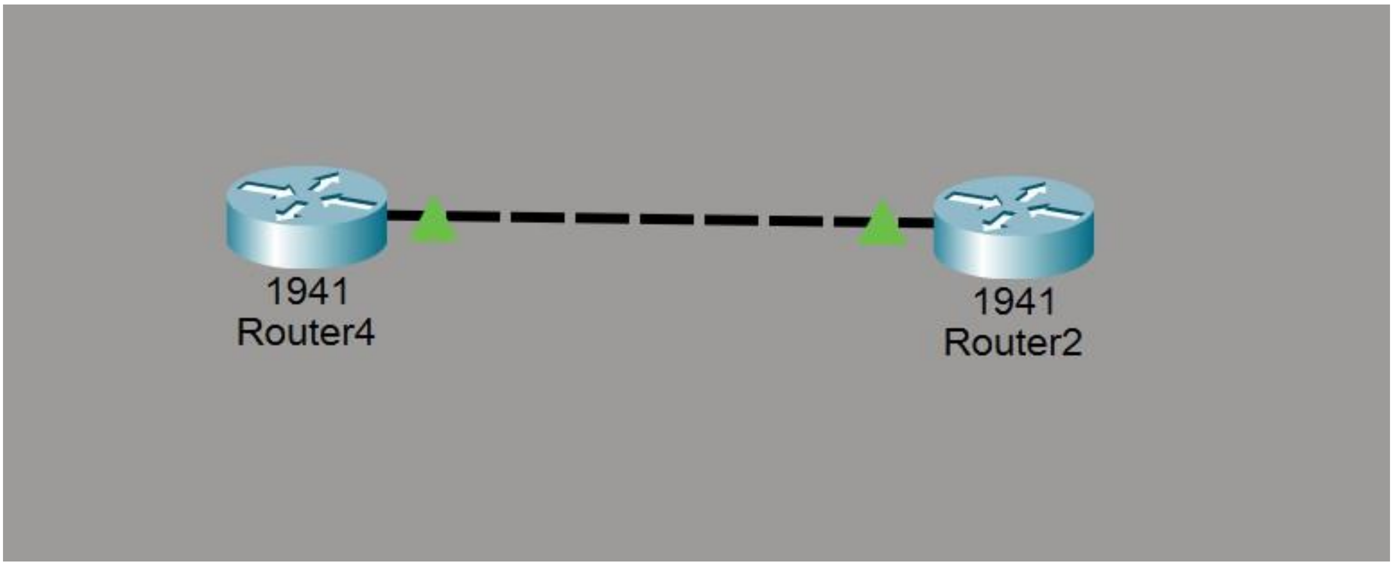


Step 14 – Router Demonstration



Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the concept of IP addressing and subnet masks.
- Configure basic network topologies in Cisco Packet Tracer.
- Use Cisco IOS CLI commands for router configuration.

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- Differentiate between Straight-through and Cross-over cables.
- Understand the purpose of the RJ-45 connector.
- Explain the working of a Layer-2 Switch.
- Configure a Router to connect different networks.
- Verify successful communication using the Ping command.
- Develop a basic understanding of computer network configuration and troubleshooting.